Personal Finance & Portfolio Model

A spreadsheet model for real decisions about Canadian registered accounts, an ETF, and loan repayment.

CONTEXT	ROLE	YEAR	TOOLS
Personal project	Modelling and analysis	2025	Microsoft Excel · Monarch Money

3	3	3
REGISTERED ACCOUNT TYPES COMPARED: TFSA, RRSP, FHSA	RETURN HORIZONS PER FUND: 1, 3 AND 5 YEARS	DECISIONS MODELLED: ACCOUNTS, FUND, REPAYMENT

01 BRIEF

Answer practical money questions with a model instead of a guess. Which mix of TFSA, RRSP and FHSA contributions comes out ahead over time? Is a Canadian dividend ETF worth its fees? How much does a different repayment schedule really change the interest paid?

It's on an engineering portfolio because it's the same discipline: define the inputs, model the system honestly, then compare scenarios on equal terms.

02 MODEL STRUCTURE

01 Registered accounts
Projected long-term growth across TFSA, RRSP and FHSA with annual contributions, compound growth, asset allocation and shifts in marginal tax bracket.

02 Fund suitability
Compared XEI, a Canadian equity dividend ETF, on management expense ratio, dividend yield and trailing 1-, 3- and 5-year total returns.

03 Repayment schedules
Built amortization models comparing alternative loan repayment schedules: principal reduction, timeline and cumulative interest under different rates and cash flows.

Unequal tax treatment — FAILURE MODE 7.A

PROBLEM	TFSA, RRSP and FHSA are taxed differently going in and coming out, so their balances can't be compared at face value. A larger RRSP balance can be worth less after tax than a smaller TFSA.
RESOLVED	Modelled each account with its own tax treatment and the shift in marginal bracket over time, so all three are compared on the same after-tax footing.

03 THE CORE ARITHMETIC

Future value, yearly C	$FV = C \cdot ((1 + r)^n - 1) / r$
Level payment on P	$M = P \cdot i / (1 - (1 + i)^{-N})$
Interest, period k	$I_k = i \cdot balance_{k-1}$
ETF net return	total return − MER

r: annual growth rate · n: years · i: periodic interest rate · N: number of payments.

04 OUTCOME

A reusable model that turns each of these questions into numbers that can be compared side by side.

05 REV. B

- Replace the single growth rate with a range of return scenarios.
- Rebuild the model in Python so scenarios can be swept automatically.